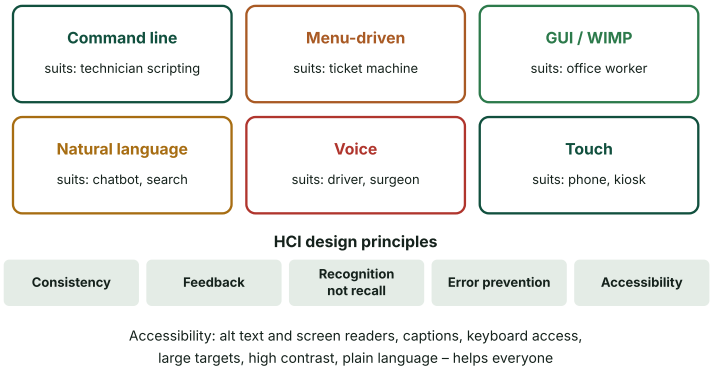
User interfaces and HCI design

FD12 FACT FILE

Specification: students should be able to

- Describe the features of a good user interface
- Describe the main features of WIMP (graphical), command line and forms dialogue interfaces
- Evaluate WIMP, command line and forms dialogue interfaces for given users and tasks
- Describe how text, sound and video can be input using personal computers, laptops, tablets and smartphones
- Describe other interfaces (voice, touch, biometric, menu-driven) and principles of HCI design, including accessibility

Key words: user interface, human–computer interaction (HCI), graphical user interface (GUI), WIMP, command line interface (CLI), switch (parameter), forms dialogue interface, menu-driven interface, voice (dialogue) interface, touch-sensitive interface, biometric interface, consistency, feedback, forgiving (undo), accessibility, speech recognition



Six types of user interface with an example user for each, and the key HCI design principles

What a user interface is and what makes it good

The user interface is the part of a system through which a person and a computer communicate: the screens, sounds, controls and input methods that let the user give instructions and receive results. A good interface is simple, with a clear layout that is easy to learn, striking a balance between offering many functions and staying uncluttered. It is intuitive and consistent, so controls work as users expect and in the same way throughout, allowing knowledge from one task to be reused in another. It is responsive, keeping up with the user's input and always giving feedback about what it is doing, such as a progress bar while a file uploads. It is forgiving, so mistakes can be reversed with undo or cancel, destructive actions are confirmed, and help is available. Finally, it is attractive, because good typography and colour make the system pleasant to use. These qualities underpin human–computer interaction (HCI) design, whose aim is to make systems effective, efficient and satisfying for the people who actually use them.

WIMP graphical user interfaces

A graphical user interface is the most common type of interface. It is often called a WIMP interface because it is built from windows, which are areas of the screen where applications and documents run; icons, which are small pictures representing programs, files and devices; menus, which list options to choose from; and a pointer, moved by a mouse, trackpad or finger, which selects items. Operating systems such as Windows, macOS, Android and iOS use graphical interfaces, usually alongside touch input. The main advantages are that a GUI is intuitive, because users recognise options instead of recalling commands, needs little training, makes it easy to exchange data between applications by copying and pasting, and offers help for beginners and keyboard shortcuts for experts. The disadvantages are that a GUI needs more memory, storage and processing power than other interfaces and a good-quality display, and it takes programmers longer to develop. Repetitive tasks can also be slower with a mouse than with a typed command or script.

Command line interfaces

A command line interface lets the user type commands at a prompt. The operating system recognises a fixed set of commands, each with a precise syntax, and many commands accept switches or parameters that change their behaviour, for example a switch that makes a copy or listing command include every subfolder. Command line interfaces were once used by everyone, but today they are mainly used by network managers, system administrators and programmers. Their advantages are that they need very little processing power, memory or display quality, and that an expert who knows the commands can work faster than with any other interface. Sequences of commands can be saved as scripts so that repetitive jobs, such as setting up dozens of user accounts, are automated. The disadvantages are that commands must be learned and typed exactly; a small mistake may mean the command fails or, worse, deletes files. There are few clues on screen about what to type, so the interface is confusing and unsuitable for novices.

Forms dialogue and other interfaces

A forms dialogue interface guides the user through a structured form using text boxes, drop-down lists, check boxes and radio buttons. It is widely used online for bookings, orders and applications. It is easy for beginners because everything is laid out logically and there are no commands to learn, it needs little processing power, and data can be validated as it is entered, reducing errors. However, a badly designed form can be slow, users can only choose from the options provided, and experienced users may find it tedious. A menu-driven interface offers a limited list of choices, often in levels, and suits kiosks and simple devices. Voice, or dialogue, interfaces let users speak commands; they are hands-free and help people who find typing difficult, but background noise, accents and words that sound alike cause errors. Touch-sensitive interfaces are intuitive and save space but make small targets hard to hit. Biometric interfaces use fingerprints or faces; they cannot be forgotten but are costly and raise privacy concerns.

Inputting text, sound and video

Different devices give users different ways of entering text, sound and video. A desktop PC usually has a full-size keyboard for fast, accurate text entry, and a scanner with optical character recognition can turn printed pages into editable text. Sound is captured through a plug-in or headset microphone, and video through a USB webcam or a capture card connected to a camera. A laptop builds these into the case: a keyboard and trackpad, a microphone array and a webcam above the screen, which makes video calls easy but gives lower quality than dedicated equipment. Tablets and smartphones rely on touch: text is entered with an on-screen keyboard, often with predictive text, swipe typing or handwriting recognition with a stylus. Speech recognition converts spoken words to text, which is quicker than typing on a small screen and helps users with dyslexia or limited mobility. Built-in microphones record voice notes, and front and rear cameras record high-resolution video that is compressed on the device so that it can be stored and shared.

HCI principles and accessibility

Human–computer interaction design applies a small number of principles. Consistency means the same layout, colours, terms and controls are used throughout, so users can predict how things work. Feedback means every action is acknowledged with a message, sound or visual change. Minimising memory load means showing options so that users recognise rather than recall them. Error prevention and recovery means validating input, confirming destructive actions and offering undo. The interface must also suit its users, tasks, environment and device. Accessibility means designing so that people with visual, hearing, motor or cognitive impairments can use the system. Examples include support for screen readers and alternative text for images, captions for video, keyboard-only navigation and large targets for people with limited dexterity, adjustable text size and strong colour contrast, never using colour as the only way of conveying meaning, and plain language with enough time to complete tasks. Accessible design helps everyone, including older users and people using a phone in bright sunlight, and supports legal duties under disability discrimination law.

T Key terms from the mark schemes

User interface
The means by which a user and a computer communicate, including what is displayed and how the user gives input.

Graphical user interface (GUI)
A user interface that uses windows, icons, menus and a pointer (WIMP) to allow the user to interact visually.
`Credited: WIMP interface`

Command line interface (CLI)
A user interface in which the user types commands in an exact syntax at a prompt for the computer to carry out.

Menu-driven interface
A user interface that presents the user with a limited list of options to choose from, often in a hierarchy.

Biometric interface
A user interface operated by measuring a physical characteristic of the user, such as a fingerprint, face or iris.

Human–computer interaction (HCI)
The study and design of how people interact with computers, aiming to make systems effective, efficient and satisfying to use.

Switch
An extra parameter typed after a command in a command line interface to modify what the command does.
Credited: Parameter / option

Forms dialogue interface
A user interface that guides the user through a structured on-screen form using text boxes, drop-down lists, check boxes and radio buttons.
Credited: Forms interface

Voice (dialogue) interface
A user interface that is controlled by spoken commands, using speech recognition.
Credited: Speech interface

Natural language interface
A user interface that accepts requests typed or spoken in ordinary everyday language.

Accessibility
Designing systems so they can be used by people with a wide range of abilities, including visual, hearing, motor and cognitive impairments.

C C# example

A tiny command interpreter in C#

```
string[] files = { "timetable.docx", "rota.xlsx" };
Console.Write("> ");
string command = Console.ReadLine() ?? "";
switch (command)
{
    case "list":
        foreach (string f in files) Console.WriteLine(f);
        break;
    case "count":
        Console.WriteLine(files.Length + " files");
        break;
    default:
        Console.WriteLine("'" + command + "' is not recognised");
        break;
}
```

Exam focus

Questions on interfaces are usually State, Describe or Explain items worth one or two marks per point, plus scenario questions asking you to recommend or evaluate an interface. Mark schemes credit the specific advantages and disadvantages listed for each interface: for example, a command line uses little memory and is fast for experts who know the commands, but commands must be typed precisely and it is unsuitable for novices. When asked why an expert might prefer a command line, link speed to their knowledge of the commands. For WIMP, name all four elements correctly: windows, icons, menus and pointer. For a forms interface, remember that validation can be carried out as data is entered. In recommendation questions, name the interface and justify it using the user's characteristics, such as expertise, disability, environment or task; vague phrases like 'easy to use' earn little unless you explain why and for whom. Extended questions need both benefits and drawbacks before a justified conclusion. Do not confuse a menu-driven interface with the menus of a GUI.

? Check yourself

1. A council is designing a self-service kiosk for paying parking fines. State four qualities its user interface should have. [4]
2. State what each letter of WIMP stands for. [2]
3. Give two advantages of a command line interface for an experienced network technician at a further education college. [2]

Answers: 1. Any four from: simple / clear, easy to learn (1) / intuitive and consistent (1) / responsive, giving feedback on status (1) / forgiving – mistakes easily reversed, help available (1) / attractive layout, typography and colour (1) / accessible (1) 2. Windows / icons / menus / pointer: all four correct (2); two or three correct (1). 3. Any two from: fast to use when the commands are known (1) / commands can be scripted to automate repetitive tasks (1) / needs little processing power or memory, so works on low-specification or remote devices (1) / gives precise, powerful control via switches/parameters (1)